



You are an expert clinical causal inference researcher, medical ML scientist, and causal representation learning specialist.

Your task is to construct the most clinically plausible latent-variable causal DAG for an irregular medical time-series dataset.

Dataset name:

PhysioNet 2012 Challenge

Official dataset description:

DATASET-DESCRIPTION-LINK = <https://physionet.org/content/challenge-2012/1.0.0/>

GLOBAL RESEARCH MISSION — READ BEFORE DOING ANYTHING

Your final goal is NOT merely to construct rule-based decision trees.

Your final goal is to construct a research-backed latent-variable causal model for PhysioNet 2012 Challenge.

The decision trees are only an implementation mechanism for converting observed measurements into binary proxy labels for latent clinical states. They are not the scientific endpoint.

The scientific endpoint is:

1. A clinically plausible latent causal ontology:
 - What are the true hidden clinical states behind the observed measurements?
 - Which latent states are meaningful causal variables?
 - Which observed variables are only noisy manifestations of these states?
2. A clinically and causally justified latent DAG:
 - How do background variables affect latent clinical states?
 - How do latent states affect each other?
 - How do latent states affect mortality?
 - How do latent states generate observed labs, vitals, and charted values?
 - How does missingness/measurement intensity arise from latent severity and clinical decision-making?
3. A causal-inference-ready graph:
 - Which variables are confounders?
 - Which variables are mediators?
 - Which variables are colliders?
 - Which variables should or should not be adjusted for when estimating effects on mortality?
 - Which latent states are outcome-proximal severity markers rather than valid adjustment variables?
4. A practical operationalization layer:
 - Only after the latent variables are selected and justified, create rule-based binary decision trees.
 - These trees produce implementable proxy labels such as LAT_SHOCK, LAT_RENAL_DYSFUNCTION, and LAT_GLOBAL_SEVERITY.
 - The trees are tools for implementation, not the main research contribution.

Priority order:

1. Research-backed latent clinical states.
2. Research-backed latent causal DAG.
3. Measurement model from latent states to observed variables.
4. Missingness and irregular-sampling model.

5. Confounding/mediation/collider analysis.
6. Rule-based decision trees for implementation.

If there is any conflict between making a simple decision tree and preserving causal correctness, preserve causal correctness.

Core objective:

Construct a clinically grounded latent causal representation of the dataset, including:

1. A set of causal latent variables representing underlying patient clinical states.
2. Formal definitions of these latent variables using available observed measurements and metadata.
3. Rule-based binary decision trees that convert observational/meta data into binary latent-variable labels.
4. A directed acyclic graph (DAG) over background variables, latent clinical states, observed measurements, measurement/missingness processes, and mortality.
5. A clear explanation of how the latent clinical states causally affect mortality.
6. A clear explanation of how the observed variables are generated as manifestations, measurements, or proxies of the latent clinical states.

Important conceptual rule:

The observed variables in the dataset are not the main causal variables. They are clinical measurements, test results, charted values, or proxies. The causal variables of interest are latent clinical states such as shock, respiratory failure, renal dysfunction, hepatic dysfunction, inflammation/sepsis burden, coagulation dysfunction, metabolic derangement, neurologic dysfunction, cardiac injury, global severity, chronic baseline risk, or other dataset-appropriate latent states.

Do not construct a naive causal graph where every lab/vital directly causes every other lab/vital. Instead, construct a causal representation in which:

- Background/static variables influence latent patient states.
- Latent patient states influence observed measurements.
- Latent patient states influence mortality.
- Observed measurements are mainly manifestations of latent physiology, not primary causal mechanisms.
- Measurement frequency and missingness may themselves be informative and influenced by latent severity, clinical decision-making, or care intensity.

Your goal is to approximate the true underlying clinical causal structure as well as possible using deep research, medical reasoning, causal inference principles, and the official dataset documentation. Do not merely produce a correlation graph.

PHASE 1 — Dataset audit

First, open and study DATASET-DESCRIPTION-LINK carefully.

Extract:

1. The complete list of available variables.
2. Which variables are static/background variables.
3. Which variables are time-varying observed measurements.
4. Which variables are outcomes, especially mortality.
5. Which variables are interventions or care-process variables, if any.
6. Units, clinical meaning, valid ranges, and measurement windows when available.
7. Known dataset collection rules and missingness patterns.
8. Whether the dataset contains ICU type, admission type, care unit, demographics, diagnoses, chart events, labs, vitals, medications, procedures, or outcomes.

Then summarize the dataset in a compact audit table.

Do not proceed to latent-variable construction until the dataset audit is complete.

PHASE 2 — Deep Research protocol and research goals

Perform deep research before constructing latent variables, decision trees, or the DAG.

The purpose of Deep Research is not to collect thresholds for decision trees.

The purpose of Deep Research is to infer the most plausible latent clinical causal structure behind the dataset.

Your research must answer the following research questions:

RQ1 — Latent-state discovery:

What hidden physiological or clinical states plausibly generate the observed measurements in PhysioNet 2012 Challenge?

Examples:

- Shock or hemodynamic instability may generate abnormal blood pressure, heart rate, lactate, urine output, and vasopressor-related signals if available.
- Respiratory failure may generate abnormal oxygenation, respiratory rate, blood gas, ventilation, and saturation signals.
- Renal dysfunction may generate abnormal creatinine, BUN, urine output, potassium, bicarbonate, and acid-base patterns.
- Inflammation/sepsis burden may generate fever/hypothermia, WBC abnormalities, lactate elevation, shock, and organ dysfunction.
- Global severity may represent multi-system physiological instability rather than a single measurement.

RQ2 — Measurement model:

For each observed variable, determine whether it is:

- a background/static cause,
- an observed proxy of a latent state,
- a treatment/intervention/care-process variable,
- a measurement/missingness-process variable,
- an outcome,
- or irrelevant/noisy for the causal model.

Do not treat observed labs and vitals as the primary causal variables unless they are true interventions.

RQ3 — Latent causal DAG:

Determine the causal relationships among:

- background variables,
- latent clinical states,
- mortality,
- observed measurements,
- and missingness/measurement processes.

The graph should represent a clinical data-generating process:

background risk → latent physiology → observed measurements and mortality.

RQ4 — Mortality mechanism:

Explain how mortality is caused through latent clinical states, not through raw measurements directly.

For example:

- chronic baseline risk → vulnerability → global severity → organ dysfunction → mortality
- acute insult → inflammation/sepsis → shock → renal/metabolic failure → mortality
- respiratory failure → hypoxemia/acidosis → global severity → mortality

Adapt these mechanisms to the actual dataset variables.

RQ5 — Missingness and irregular sampling:

Because PhysioNet 2012 Challenge is an irregular clinical time-series dataset, research whether missingness is likely:

- MCAR,
- MAR,

- MNAR,
- or caused by care intensity / clinician suspicion / latent severity.

Treat measurement frequency and missingness as potentially informative, not as random noise.

RQ6 — Operational proxy construction:

Only after RQ1–RQ5 are answered, construct binary rule-based decision trees for the approved latent variables.

The decision trees should approximate the latent states using available observed variables, but they must not replace the causal DAG.

Research sources to use:

1. DATASET-DESCRIPTION-LINK.
2. Dataset-specific papers, benchmark papers, and preprocessing conventions.
3. ICU severity literature: SOFA, APACHE, SAPS, sepsis, shock, organ failure, respiratory failure, renal failure, coagulation, hepatic dysfunction, neurologic dysfunction, metabolic derangement.
4. Causal inference literature: SCMs, DAGs, latent variables, backdoor adjustment, mediation, colliders, potential outcomes.
5. Causal representation learning literature: observed variables as functions/proxies of latent causal variables.
6. Clinical EHR/time-series literature: irregular sampling, informative missingness, measurement intensity, care processes.
7. Existing clinical causal graphs or phenotype networks, if available.

For every proposed latent variable, decision rule, and DAG edge, provide:

- clinical mechanism,
- dataset support,
- causal direction,
- whether it risks leakage,
- whether it is a confounder, mediator, collider, or measurement proxy,
- confidence level,
- and whether it should be kept, optional, or rejected.

Do not proceed from research to final DAG until you can explain why the selected latent variables are the right causal level for this dataset.

PHASE 3 — Core causal ontology

Use the following node types:

A. Background/static variables

Examples:

- Age
- Sex/gender
- Ethnicity/race, if available
- Height, weight, BMI
- ICU type, admission type, first care unit
- Chronic disease burden or baseline vulnerability, if derivable

Background variables may cause latent clinical states and mortality risk, but they are generally not caused by ICU measurements.

B. Latent clinical state variables

These are the main causal variables.

Possible candidate latent states include, but are not limited to:

- Chronic baseline risk / chronic burden
- Acute insult / acute physiological stress
- Global severity of illness
- Circulatory shock / hemodynamic instability

- Respiratory failure / pulmonary dysfunction
- Renal dysfunction / kidney injury
- Hepatic dysfunction / liver injury
- Coagulation or hematologic dysfunction
- Inflammation / infection / sepsis burden
- Neurologic dysfunction
- Cardiac injury / cardiac strain
- Metabolic derangement / acid-base-electrolyte disorder
- Nutritional, fluid balance, or volume-status burden, if supported by variables
- Treatment/care intensity, if the dataset contains care-process variables

Do not force this list. Adapt the latent variables to the actual variables in PhysioNet 2012 Challenge.

Each latent variable must have:

1. A clinical definition.
2. A causal interpretation.
3. Observable indicators from the dataset.
4. Suggested proxy rule or thresholding logic.
5. Expected direction of abnormality.
6. Confidence level.
7. Notes on possible ambiguity.

C. Observed measurement variables

These include labs, vitals, charted measurements, and test results.

Treat these as manifestations or noisy measurements of latent states, for example:

- Shock may manifest through low blood pressure, high lactate, tachycardia, vasopressor use if available.
- Respiratory failure may manifest through PaO₂, FiO₂, oxygen saturation, respiratory rate, ventilation variables if available.
- Renal dysfunction may manifest through creatinine, BUN, urine output, potassium, bicarbonate.
- Hepatic dysfunction may manifest through bilirubin, AST, ALT, albumin.
- Coagulation dysfunction may manifest through platelets, INR/PT/PTT if available.
- Inflammation/infection may manifest through WBC, temperature, lactate, cultures/antibiotics if available.
- Cardiac injury may manifest through troponin, heart rate, blood pressure, ECG-related variables if available.
- Metabolic derangement may manifest through glucose, bicarbonate, pH, electrolytes, anion gap if derivable.

An observed variable may be an indicator of more than one latent state. Allow multi-parent measurement edges when clinically justified.

D. Mortality outcome

Mortality should be treated as an outcome node.

Latent states may cause mortality directly.

Background variables may affect mortality directly or indirectly through latent states.

Observed measurements should generally not directly cause mortality unless the variable is truly an intervention or treatment exposure. Labs and vitals usually indicate the severity state rather than causing death themselves.

E. Measurement and missingness process

Because PhysioNet 2012 Challenge is an irregular medical dataset, include a measurement process if relevant:

- Measurement intensity
- Test ordering process
- Missingness indicator
- Observation availability
- Care intensity

These may be influenced by latent severity, clinician suspicion, ICU type, care unit, admission type, or background risk.

Do not assume missingness is MCAR.

Discuss whether missingness is likely MAR or MNAR.

If missingness is clinically informative, include it as a separate observation-process node, not as a biological causal state.

PHASE 4 — Latent variable selection and definition

Before constructing the DAG, define the latent variables that will appear in the DAG.

This phase must happen before DAG construction.

Your task in this phase:

1. Propose candidate latent clinical variables based on the dataset audit.
2. Map available observed and metadata variables to candidate latent variables.
3. Remove candidate latent variables that are not sufficiently supported by the dataset.
4. Merge overlapping latent variables if the dataset cannot distinguish them reliably.
5. Mark weakly measured but clinically important latent variables as optional.
6. Finalize the latent variable set before constructing causal edges.

For every latent variable, define it operationally from observed data.

For each latent variable produce:

1. Name
2. Clinical meaning
3. Why it is latent rather than directly observed
4. Observable indicators
5. Metadata indicators, if relevant
6. Suggested binary definition
7. Suggested continuous severity score, if useful
8. Thresholds or quantile-based fallback rules
9. Missingness handling
10. Directionality: higher value means worse/better?
11. Whether this latent variable should be used as:
 - prediction target,
 - confounder,
 - mediator,
 - outcome-proximal severity marker,
 - treatment-effect modifier.
12. Confidence: high / medium / low.
13. Whether to keep, merge, make optional, or reject.

Definitions must be clinically meaningful but also implementable on PhysioNet 2012 Challenge.

Do not define latent variables from a single measurement unless unavoidable.

Prefer multiple indicators and majority-vote or score-based definitions.

If the dataset lacks sufficient indicators for a latent state, either:

- remove the latent variable,
- mark it as weakly measured,
- merge it with another latent variable,
- or define it as optional.

Do not use mortality or post-outcome information to define any latent variable.

Do not allow outcome leakage.

Do not use future measurements if the latent label is intended to represent an earlier clinical state.

At the end of this phase, provide a final approved latent-variable list.

Only these approved latent variables may be used as latent nodes in the DAG.

IMPORTANT — Decision trees are not the final goal

The rule-based binary decision trees are only an operational layer.

They are used to create binary proxy labels for latent variables so that the thesis pipeline can generate latent_tags.csv and use those labels in downstream modeling.

Do not optimize the entire answer around the decision trees.

A correct answer with clinically meaningful latent variables and a strong DAG is better than a large collection of simplistic decision trees.

The decision trees must serve the causal model, not define the causal model.

Before writing any tree, explicitly state:

- which latent clinical state the tree approximates,
- why this state belongs in the DAG,
- which observed variables are manifestations of that state,
- and how the proxy label may differ from the true unobserved clinical state.

PHASE 5 — Rule-based binary decision trees for latent labels

After the latent-variable set is finalized, construct a rule-based binary decision tree for each approved latent variable.

Use the uploaded image decision-tree-example.png as the structural template.

The desired structure is:

1. Define several clinically meaningful domain indicators.
2. Each domain indicator is a Boolean condition based on observed measurements or metadata.
3. Combine domain indicators into a score:
score = sum(domain indicators)
4. Apply a threshold:
if score >= THRESHOLD:
LATENT_VARIABLE = 1
else:
LATENT_VARIABLE = 0

Example structure from decision-tree-example.png :

- Circulatory domain:
MAP_first < 70 OR SysABP_first <= 100
- Neurologic domain:
GCS_first < 15
- Respiratory domain:
RespRate_first >= 22 OR SaO2_first < 92
- Metabolic domain:
Lactate_first > 2 OR pH_first < 7.3 OR HCO3_first < 18
- Renal domain:
Creatinine_first >= 2
- score = sum(domain indicators)
- score >= 2 gives Severity = 1
- otherwise Severity = 0

For each latent variable, create a similar binary decision tree.

Requirements:

1. The output of every tree must be binary: 0 or 1.
2. The rules must use only observed variables and metadata available in PhysioNet 2012 Challenge.
3. Use clinically interpretable thresholds when possible.

4. If standard clinical thresholds are not available or not dataset-appropriate, propose quantile-based fallback thresholds.
5. Each tree should use multiple indicators when possible.
6. A tree may use first value, worst value, mean value, slope, variability, measurement frequency, or missingness indicators, but you must justify each choice.
7. Avoid using mortality, survival time, discharge disposition, or outcome-derived variables.
8. Avoid using future values if the latent variable is meant to represent early-state physiology.
9. Use missingness explicitly when clinically informative, but do not let missingness dominate the latent definition unless justified.
10. Mark rules as high / medium / low confidence.

For each decision tree, provide:

1. Latent variable name.
2. Clinical interpretation of label 1.
3. Input variables.
4. Domain indicators.
5. Scoring function.
6. Binary threshold.
7. Missingness policy.
8. Pseudocode.
9. A compact Mermaid flowchart.
10. Clinical rationale.
11. Potential false positives and false negatives.
12. Confidence level.

The decision trees should be designed so they can be directly implemented in code to generate a latent_tags.csv file.

Use naming convention:

LAT_<LATENT_NAME> = 0/1

For example:

LAT_SHOCK = 1 means the patient has evidence of circulatory shock or clinically meaningful hemodynamic instability.

LAT_RENAL_DYSFUNCTION = 1 means the patient has evidence of acute or clinically meaningful renal dysfunction.

LAT_GLOBAL_SEVERITY = 1 means the patient has evidence of multi-domain severity according to the score-based tree.

After creating the decision trees, verify:

- Every approved latent variable has a binary tree.
- Every tree uses only allowed variables.
- No tree uses mortality or future outcome information.
- Every threshold is clinically or statistically justified.
- Every tree can be implemented on irregular time-series data.

Only after this verification, proceed to DAG construction.

PHASE 6 — DAG construction rules

Construct the DAG only after:

1. The dataset variables were audited.
2. The latent variables were selected and finalized.
3. The binary decision trees were constructed and validated.

The final graph must be a DAG.

Rules:

1. No directed cycles.
2. No edge from mortality back to clinical states.
3. No edge from future measurements to earlier latent states.
4. No automatic observed-variable-to-observed-variable causal chains unless there is a true clinical mechanism.
5. Prefer latent-state-mediated explanations over direct lab-to-lab causation.
6. Use temporal and clinical ordering:
 - Background risk precedes acute illness.
 - Acute insult precedes or contributes to global severity.
 - Global severity may contribute to organ dysfunction.
 - Organ dysfunction contributes to mortality.
 - Organ dysfunction produces observed abnormal measurements.
7. Allow clinically justified latent-to-latent edges, such as:
 - Chronic risk → acute vulnerability
 - Acute insult → global severity
 - Global severity → shock
 - Shock → renal dysfunction
 - Shock → hepatic dysfunction
 - Shock → metabolic derangement
 - Inflammation/sepsis → shock
 - Inflammation/sepsis → coagulation dysfunction
 - Respiratory failure → metabolic derangement
 - Renal dysfunction → metabolic derangement
 - Cardiac injury → shock
8. Avoid over-dense DAGs. Use sparse, clinically meaningful edges.
9. Each edge must have a mechanism and confidence score.
10. If an edge is plausible but uncertain, mark it as optional or low-confidence.
11. Do not confuse measurement with intervention.
12. Do not confuse association with causation.
13. Do not use the mortality label to define latent variables in a way that leaks the outcome.
14. Observed variables should usually be children of latent states, not parents of latent states.
15. Binary latent labels generated by the decision trees are operational proxies for latent states, not the biological states themselves. Make this distinction explicit.

DAG construction order:

1. Add background/static nodes.
2. Add approved latent clinical-state nodes.
3. Add mortality outcome node.
4. Add observed measurement nodes as manifestations of latent states.
5. Add measurement/missingness process nodes if needed.
6. Add background-to-latent edges.
7. Add latent-to-latent edges.
8. Add latent-to-mortality edges.
9. Add latent-to-observed measurement edges.
10. Add latent/background-to-measurement-process edges.
11. Check acyclicity.
12. Check clinical plausibility.
13. Check whether the graph is sparse enough.

PHASE 7 — Iterative validation procedure

Do not produce the final answer in one pass.

Use the following internal process:

Round 1:

Create an initial list of candidate latent variables from the official dataset variables.

Round 2:

Map every observed variable to one or more latent clinical states. Identify unassigned variables and decide whether they are irrelevant, background, outcome, treatment, or measurement-process variables.

Round 3:

Finalize the latent-variable set. Remove, merge, or mark optional latent variables that are poorly supported.

Round 4:

Construct rule-based binary decision trees for every approved latent variable using observed and metadata variables.

Round 5:

Validate the decision trees:

- Do the rules use only available variables?
- Are thresholds clinically meaningful?
- Is missingness handled correctly?
- Is there outcome leakage?
- Are the trees implementable on irregular time-series data?
- Are false positives/false negatives acceptable?

Round 6:

Construct a first-pass latent DAG using only approved latent variables. Check for cycles, clinically implausible edges, and over-dense structure.

Round 7:

Research again. Compare the graph to known clinical severity frameworks and EHR causal/phenotype-network literature. Revise latent variables, decision rules, and edges if needed.

Round 8:

Perform adversarial critique:

- Which latent variables are not truly identifiable from the dataset?
- Which decision-tree rules are too weak or too dependent on a single measurement?
- Which edges are only correlations?
- Which observed variables were incorrectly treated as causes?
- Which latent variables are poorly measured?
- Which missingness patterns may bias definitions?
- Which edges create collider bias if adjusted for?
- Which variables should not be adjusted for when estimating effects on mortality?
- Which variables are mediators rather than confounders?
- Which latent definitions risk outcome leakage?

Only after these rounds, produce the final answer.

PHASE 8 — Required final output

Your final answer must contain the following sections.

1. Dataset audit

A table with:

- Variable name
- Variable type: background / observed time-varying / outcome / treatment / measurement-process
- Clinical meaning

- Unit if available
 - Notes
2. Candidate latent variables
- A table with:
- Candidate latent variable name
 - Clinical meaning
 - Supporting observed variables
 - Dataset support level
 - Keep / merge / optional / reject
 - Rationale
3. Final approved latent variables
- A table with:
- Latent variable name
 - Clinical meaning
 - Observed indicators
 - Suggested definition
 - Missingness strategy
 - Confidence: high / medium / low
 - Rationale
4. Rule-based binary decision trees
- For every approved latent variable, provide:
- Latent variable name
 - Binary label definition
 - Input variables
 - Domain indicators
 - Score formula
 - Binary threshold
 - Missingness policy
 - Pseudocode
 - Mermaid flowchart
 - Clinical rationale
 - False positive risks
 - False negative risks
 - Confidence level

The structure should resemble decision-tree-example.png:

domain indicators → score = sum(domain indicators) → threshold → binary label.

5. Observed-to-latent measurement model
- A table with:
- Observed variable
 - Parent latent variable or variables
 - Direction of abnormality
 - Whether it is specific or non-specific
 - Whether it is used in a binary decision tree
 - Notes
6. Latent causal DAG edge list
- A table with:
- Source node
 - Target node
 - Edge type: background-to-latent / latent-to-latent / latent-to-observed / latent-to-mortality / measurement-process

- Clinical mechanism
- Confidence
- Source or reasoning basis
- Keep / optional / reject

7. Mortality mechanism summary

Explain the main causal paths to mortality, for example:

- Chronic baseline risk → global severity → organ dysfunction → mortality
- Acute insult → shock → renal/metabolic failure → mortality
- Inflammation/sepsis → shock/coagulation dysfunction → mortality
- Respiratory failure → hypoxemia/acidosis → mortality

Adapt these paths to the actual dataset and final DAG.

8. Missingness and irregular sampling analysis

Discuss:

- Which variables are likely MNAR or MAR.
- Whether missingness should be represented as a separate node.
- Whether measurement frequency can act as a proxy for severity or care intensity.
- How missingness should be handled when defining latent variables.
- How missingness enters or does not enter the binary decision trees.

9. Backdoor/confounding implications

For the purpose of estimating causal effects on mortality:

- Identify likely confounders.
- Identify likely mediators.
- Identify colliders to avoid conditioning on.
- Explain which latent variables should be used for adjustment and which should not, depending on the treatment/exposure of interest.
- Explain whether the binary latent tags are safe adjustment variables or whether they may be descendants/mediators.

10. Graph specification

Provide the final DAG as:

- Adjacency list.
- Mermaid graph.
- Python NetworkX edge list.

Use clear node prefixes:

- BG_ for background variables
- LAT_ for latent variables
- OBS_ for observed measurements
- MISS_ for missingness/measurement-process variables
- OUT_ for mortality outcome

11. Decision-tree implementation specification

Provide a code-oriented specification for generating latent_tags.csv:

- Required input columns.
- Derived summary features, such as first, worst, mean, slope, count, missingness.
- One function per latent variable.
- Expected binary output column names.
- Handling of missing variables.
- Unit assumptions.
- Threshold table.
- Validation checks.

12. Rejected alternatives

List important variables, latent definitions, decision rules, or edges that were considered but rejected, with reasons.

13. Final confidence and limitations

State:

- Which latent variables are strongest.
- Which are weakest.
- Which decision trees are most reliable.
- Which decision trees are most uncertain.
- Which edges are most uncertain.
- What additional data would improve identifiability.
- Why the graph is a clinically plausible causal DAG rather than a proven ground-truth DAG.

PHASE 9 — Style and rigor requirements

Be rigorous and skeptical.

Every final latent variable must be selected before the DAG is constructed.

Every final latent variable must have a binary decision tree.

Every final decision tree must be implementable in code.

Every final edge must be justified.

Every latent variable must be clinically interpretable.

Every observed variable must be categorized.

Do not hallucinate variables not present in the dataset.

Do not overclaim certainty.

Do not present the graph as statistically proven from observational data alone.

Do not ignore informative missingness.

Do not treat labs/vitals as primary causal states.

Do not create a dense graph just because variables correlate.

Do not define latent variables after creating the DAG.

Do not construct the DAG before the latent variables and binary decision trees are stable.

The final output should be detailed enough to implement in code:

- latent tag construction,
- rule-based binary decision trees,
- latent_tags.csv generation,
- DAG construction,
- confounder selection,
- mortality-effect estimation,
- sensitivity analysis.

Begin now by opening <https://physionet.org/content/challenge-2012/1.0.0/> and auditing the dataset variables.

Below is a clinically plausible latent-variable causal model for PhysioNet 2012 in which ICU labs and vitals are treated mainly as manifestations of a smaller set of hidden physiologic states, not as the primary causal variables themselves. The strongest latent layer supported by the challenge files is organ-system dysfunction plus a downstream multisystem severity construct; diagnosis-level states such as sepsis source, chronic comorbidity burden, or specific cardiac syndromes are only weakly observed because the challenge files do not provide diagnoses, medications, procedures, cultures, or admission-type detail beyond ICU type and mechanical ventilation.^{[1][2]}

Dataset audit

PhysioNet 2012 contains 12,000 adult ICU stays drawn from MIMIC-II, restricted to first ICU stays lasting at least 48 hours, with the prediction window being the first 48 ICU hours. The official files provide six admission descriptors, 37 time-series variables, time stamps in elapsed hours:minutes since ICU admission, and separate outcome descriptors; valid values are non-negative and -1 denotes missing or unknown.^{[2][1]}

The challenge files include one obvious care-process/treatment variable, MechVent, and repeated Weight, which can act as a fluid-balance proxy, but they do not include medication administrations, vasopressors, antibiotics, procedures, diagnoses, or culture results. That absence is the main reason to model diagnosis-level constructs such as infection source or chronic comorbidity as weak or unidentifiable latent causes rather than hard-labeled nodes.^[^2]

Background and outcomes

Variable	Type	Clinical meaning	Unit / coding	Notes
RecordID	Background identifier	Unique ICU-stay identifier [^2]	Integer [^2]	Not a causal variable.
Age	Background	Age at ICU admission [^2]	Years [^2]	Adult cohort only; age ≥ 16 years in source extraction [^1].
Gender	Background	Sex coded at admission [^2]	0=female, 1=male [^2]	Use as background risk, not a latent state.
Height	Background	Admission height [^2]	cm [^2]	Can support body-size normalization.
ICUType	Background	ICU service type [^2]	1=CCU, 2=CSRU, 3=MICU, 4=SICU [^2]	Important for case-mix and measurement intensity.
Weight (admission)	Background	Initial body weight [^2]	kg [^2]	Needed for urine normalization if available.
SAPS-I	Outcome-related descriptor	Severity score supplied separately [^2]	Score [^2]	Must not be used to define latent tags because it is an outcome-side summary.
SOFA	Outcome-related descriptor	Organ-failure score supplied separately [^2]	Score [^2]	Useful for validation only, not latent-label construction.
Length of stay	Outcome-related descriptor	Hospital length of stay after ICU admission [^2]	Days [^2]	Post-baseline; leakage if used in tag construction.
Survival	Outcome-related descriptor	Days from ICU admission to death, or -1 if alive [^2]	Days [^2]	Post-outcome descriptor; never use in latent definitions.
In-hospital death	Outcome	Mortality endpoint [^2]	0/1 [^2]	Final outcome node.

Time-varying observed variables

Variable	Audit type	Clinical meaning	Unit	Notes
Albumin	Observed time-varying	Serum albumin [^2]	g/dL [^2]	Available in about 40.5% of patients in first 48h, so useful but sparse [^1].
ALP	Observed time-varying	Alkaline phosphatase [^2]	IU/L [^2]	Hepatobiliary marker; about 42.4% availability [^1].
ALT	Observed time-varying	Alanine transaminase [^2]	IU/L [^2]	Hepatic injury marker; about 43.4% availability [^1].
AST	Observed time-varying	Aspartate transaminase [^2]	IU/L [^2]	Hepatic injury marker; about 43.4% availability [^1].
Bilirubin	Observed time-varying	Serum bilirubin [^2]	mg/dL [^2]	Hepatic/cholestatic marker; about 43.4% availability [^1].
BUN	Observed time-varying	Blood urea nitrogen [^2]	mg/dL [^2]	Common renal marker; about 98.4% availability [^1].
Cholesterol	Observed time-varying	Serum cholesterol [^2]	mg/dL [^2]	Very sparse, about 7.9%; low priority for causal ontology [^1].
Creatinine	Observed time-varying	Serum creatinine [^2]	mg/dL [^2]	Core renal marker; about 98.4% availability [^1].
DiasABP	Observed time-varying	Invasive diastolic arterial BP [^2]	mmHg [^2]	High availability; hemodynamic indicator [^1].
FiO2	Observed time-varying	Fraction of inspired oxygen [^2]	0-1 [^2]	Respiratory severity and care-intensity marker; about 67.6% availability [^1].
GCS	Observed time-varying	Glasgow Coma Score [^2]	3-15 [^2]	Core neurologic indicator; about 98.4% availability [^1].
Glucose	Observed time-varying	Serum glucose [^2]	mg/dL [^2]	Metabolic stress marker; about 97.5% availability [^1].
HCO3	Observed time-varying	Serum bicarbonate [^2]	mmol/L [^2]	Acid-base marker; about 98.2% availability [^1].
HCT	Observed time-varying	Hematocrit [^2]	% [^2]	Blood-loss/hemodilution severity proxy; about 98.4% availability [^1].
HR	Observed time-varying	Heart rate [^2]	bpm [^2]	Core hemodynamic stress marker; about 98.4% availability [^1].

Variable	Audit type	Clinical meaning	Unit	Notes
K	Observed time-varying	Serum potassium [^2]	mEq/L [^2]	Renal/metabolic marker; about 97.9% availability [^1].
Lactate	Observed time-varying	Serum lactate [^2]	mmol/L [^2]	Hypoperfusion/metabolic marker; about 54.8% availability [^1].
Mg	Observed time-varying	Serum magnesium [^2]	mmol/L [^2]	Metabolic/renal marker; about 97.5% availability [^1].
MAP	Observed time-varying	Invasive mean arterial pressure [^2]	mmHg [^2]	Core shock marker; about 98.4% availability [^1].
MechVent	Treatment / care-process	Mechanical ventilation indicator [^2]	0/1 [^2]	Only explicit treatment-like variable in files; about 63.1% availability [^1].
Na	Observed time-varying	Serum sodium [^2]	mEq/L [^2]	Metabolic/renal marker; about 98.2% availability [^1].
NIDiasABP	Observed time-varying	Non-invasive diastolic BP [^2]	mmHg [^2]	Backup hemodynamic measure; about 87.3% availability [^1].
NIMAP	Observed time-varying	Non-invasive mean arterial pressure [^2]	mmHg [^2]	Backup shock measure; about 87.2% availability [^1].
NISysABP	Observed time-varying	Non-invasive systolic BP [^2]	mmHg [^2]	Backup shock measure; about 87.6% availability [^1].
PaCO2	Observed time-varying	Arterial carbon dioxide pressure [^2]	mmHg [^2]	ABG marker; about 75.4% availability [^1].
PaO2	Observed time-varying	Arterial oxygen pressure [^2]	mmHg [^2]	ABG oxygenation marker; about 75.4% availability [^1].
pH	Observed time-varying	Arterial pH [^2]	0-14 [^2]	Acid-base marker; about 75.9% availability [^1].
Platelets	Observed time-varying	Platelet count [^2]	cells/nL [^2]	Hematologic severity proxy; about 98.3% availability [^1].
RespRate	Observed time-varying	Respiratory rate [^2]	bpm [^2]	Surprisingly sparse at about 27.7% [^1].
SaO2	Observed time-varying	Oxygen saturation [^2]	% [^2]	Oxygenation marker; about 44.7% availability [^1].
SysABP	Observed time-varying	Invasive systolic arterial BP [^2]	mmHg [^2]	Core shock marker; about 98.4% availability [^1].
Temp	Observed time-varying	Temperature [^2]	Celsius [^2]	Inflammatory stress proxy; about 98.4% availability [^1].
TropI	Observed time-varying	Troponin-I [^2]	ug/L [^2]	Extremely sparse at about 4.7%; weak support for cardiac-injury latent [^1].
TropT	Observed time-varying	Troponin-T [^2]	ug/L [^2]	Sparse at about 21.9%; still weak for broad cardiac latent [^1].
Urine	Observed time-varying	Urine output [^2]	mL [^2]	Core renal and perfusion marker; about 97.4% availability [^1].
WBC	Observed time-varying	White blood cell count [^2]	cells/nL [^2]	Inflammatory stress proxy; about 98.2% availability [^1].
Weight (time series)	Observed time-varying	Repeated body weight [^2]	kg [^2]	May reflect fluid status and care process; about 67.7% availability [^1].

Latent ontology

The official dataset is best represented as background risk → latent physiologic states → observed measurements and mortality, with missingness partly driven by severity and clinician ordering rather than randomness. That framing is consistent with the challenge design, which emphasizes irregular measurements, dynamic changes over 48 hours, and the risk that some observed vitals reflect intervention as much as genuine physiologic state.^{[1][2]}

Candidate latent variables

Candidate latent	Clinical meaning	Supporting observed variables	Dataset support	Decision
LAT_SHOCK	Hemodynamic instability / impaired perfusion	MAP, SysABP, NIMAP, NISysABP, HR, Lactate, Urine, pH, HCO3 [^{^2}]	Strong; core variables mostly common, lactate moderate [^{^1}]	Keep
LAT_RESP_FAILURE	Hypoxemic or ventilatory respiratory failure	PaO2, FiO2, SaO2, PaCO2, RespRate, MechVent, pH [^{^2}]	Strong-moderate; ABG support reasonable, MechVent explicit [^{^1}]	Keep
LAT_RENAL_DYSFUNCTION	AKI / kidney dysfunction	Creatinine, BUN, Urine, K, HCO3 [^{^2}]	Strong [^{^1}]	Keep
LAT_HEPATIC_DYSFUNCTION	Liver injury / cholestasis / impaired synthetic reserve	Bilirubin, AST, ALT, ALP, Albumin [^{^2}]	Moderate because liver labs are only about 40–43% available [^{^1}]	Keep
LAT_NEURO_DYSFUNCTION	Depressed consciousness / neurologic failure	GCS [^{^2}]	Strong single-indicator support because GCS is common [^{^1}]	Keep, but acknowledge near-direct measurement
LAT_METABOLIC_DERANGEMENT	Acidosis, hyperlactatemia, glucose/electrolyte disorder	pH, HCO3, Lactate, Glucose, Na, K, Mg [^{^2}]	Strong [^{^1}]	Keep
LAT_GLOBAL_SEVERITY	Multisystem physiologic instability	Aggregate of organ-state indicators plus broad vitals/labs [^{^2}]	Strong as a higher-order latent, not directly observed [^{^1}]	Keep
LAT_HEMATOLOGIC_DYSFUNCTION	Thrombocytopenia / bleeding-related dysfunction	Platelets, HCT [^{^2}]	Moderate but not rich enough for a robust standalone latent [^{^1}]	Merge into global severity / optional
LAT_SEPSIS_INFLAMMATION	Infection/inflammatory burden	Temp, WBC, Lactate, BP [^{^2}]	Weak because no cultures, antibiotics, source, or diagnosis [^{^2}]	Reject as final node
LAT_CARDIAC_INJURY	Myocardial injury / strain	TropI, TropT, HR, BP [^{^2}]	Weak because troponins are sparse [^{^1}]	Reject as final node
LAT_CHRONIC_BURDEN	Baseline chronic disease vulnerability	Age, ICUtype, Weight, Height [^{^2}]	Weak without diagnoses/comorbidities [^{^2}]	Represent with observed BG nodes instead

Final approved latent variables

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness strategy	Confidence	
LAT_SHOCK	Hemodynamic instability with hypoperfusion	BP, HR, lactate, urine are manifestations, not the state itself	MAP/SBP, HR, Lactate, Urine, pH/HCO3	Label 1 if at least 2 shock-domain indicators fire	Require at least 2 observed domains or return NA	High	Str su ch vit ch [1].
LAT_RESP_FAILURE	Hypoxemic and/or hypercapnic respiratory failure	Oxygenation and ventilation measures are outputs of pulmonary dysfunction and support	PaO2/FiO2, SaO2, PaCO2, RespRate, MechVent, pH	Label 1 if at least 2 respiratory-domain indicators fire	Use PF ratio when both PaO2 and FiO2 exist; otherwise fall back to SaO2, PaCO2, MechVent	High-medium	Clk to res org str [2]
LAT_RENAL_DYSFUNCTION	Acute renal dysfunction / AKI-like state	Creatinine and urine are proxies of renal filtration and output	Creatinine, BUN, Urine, K, HCO3	Label 1 if at least 2 renal-domain indicators fire	If urine missing, rely on creatinine/BUN/electrolytes; if labs missing, keep NA	High	Re me coi clii col

Latent variable	Clinical meaning	Why latent	Observed indicators	Suggested binary definition	Missingness strategy	Confidence	
LAT_HEPATIC_DYSFUNCTION	Hepatocellular injury or cholestatic dysfunction	Liver damage is not directly observed; labs are partial manifestations	Bilirubin, AST, ALT, ALP, Albumin	Label 1 if at least 2 hepatic-domain indicators fire	Sparse lab burden means frequent NA; never infer from albumin alone	Medium	Plz lim sp tes
LAT_NEURO_DYSFUNCTION	Depressed consciousness / severe encephalopathy proxy	True neurologic failure is broader than charted GCS	GCS	Label 1 if GCS-based severity indicator fires; use one strong indicator because dataset lacks richer neuro data	If all GCS missing, NA	Medium	Ne me lim str av: ne GC
LAT_METABOLIC_DERANGEMENT	Systemic acid-base / electrolyte / glucose derangement	pH, bicarbonate, lactate, glucose, sodium, potassium are manifestations of metabolic stress	pH, HCO3, Lactate, Glucose, Na, K, Mg	Label 1 if at least 2 metabolic-domain indicators fire	Use available domains only if at least 2 domains observed	High-medium	Go an fro org no
LAT_GLOBAL_SEVERITY	Multiorgan physiologic instability	Not directly measured; represented by simultaneous derangement across systems	Outputs of the other latent tags plus broad instability	Label 1 if at least 2 organ/system latent tags are positive, or 1 plus extreme GCS/PaO2-FiO2/lactate abnormality	Compute only after other tags	High-medium	Be prc lat su th [2]

Proxy rules and measurement model

For operational tags, I would index each patient on a prespecified early window, preferably the first 24 hours for causal analyses and optionally 0–48 hours for challenge-style benchmarking, because the files contain repeated measurements throughout the first 48 hours and later values can mix evolving disease with treatment response. The rules below are therefore explicit approximations of latent states, not claims that the true hidden state is fully identified.^{[1][2]}

Rule-based binary trees

1) LAT_SHOCK

- Binary label: 1 = evidence of clinically meaningful circulatory shock / hemodynamic instability.
- Inputs: MAP, SysABP, NIMAP, NISysABP, HR, Lactate, Urine, pH, HCO3, admission Weight.
- Domain indicators:
 - $D1_{\text{hypotension}} = (\min(\text{MAP}, \text{NIMAP}) < 65) \text{ OR } (\min(\text{SysABP}, \text{NISysABP}) < 90)$
 - $D2_{\text{hypoperfusion}} = (\max(\text{Lactate}) > 2.2) \text{ OR } (\min(\text{pH}) < 7.30) \text{ OR } (\min(\text{HCO}_3) < 18)$
 - $D3_{\text{compensation}} = \max(\text{HR}) > 110$
 - $D4_{\text{oliguria}} = \text{urine}_{24\text{h}} < 500 \text{ OR } \text{urine_mlkgph} < 0.5$
- Score: $\text{shock_score} = D1 + D2 + D3 + D4$
- Threshold: $\text{LAT_SHOCK} = 1 \text{ if } \text{shock_score} \geq 2 \text{ else } 0$
- Missingness: if fewer than 2 domains are observable, set LAT_SHOCK = NA; if lactate missing, do not auto-impute normal.
- False positives: pain/anxiety tachycardia, chronic low BP, hemorrhage without septic/cardiogenic shock.
- False negatives: vasopressor-corrected shock, occult hypoperfusion with sparse lactate.

Pseudocode:

```
shock_score = hypotension + hypoperfusion + compensation + oliguria
LAT_SHOCK = int(shock_score >= 2)
```

```
flowchart TD
A[Hypotension domain] --> S
B[Hypoperfusion domain] --> S
C[Tachycardia domain] --> S
D[Oliguria domain] --> S
S[shock_score = sum(domains)] --> T{>= 2?}
T -->|Yes| Y[LAT_SHOCK = 1]
T -->|No| N[LAT_SHOCK = 0]
```

2) LAT_RESP_FAILURE

- Binary label: 1 = evidence of respiratory failure.
- Inputs: PaO₂, FiO₂, SaO₂, PaCO₂, RespRate, MechVent, pH.
- Domain indicators:
 - D1_oxygenation = (min(PaO₂/FiO₂) < 300) OR (min(SaO₂) < 92)
 - D2_vent_support = any(MechVent == 1)
 - D3_ventilatory = (max(PaCO₂) > 50) OR ((min(pH) < 7.30) AND (max(PaCO₂) > 45))
 - D4_respiratory_distress = max(RespRate) > 24
- Score: resp_score = D1 + D2 + D3 + D4
- Threshold: LAT_RESP_FAILURE = 1 if resp_score >= 2 else 0
- Missingness: require at least 2 respiratory domains observed; if PaO₂ or FiO₂ missing, use SaO₂ and MechVent rather than imputing PF ratio.
- False positives: elective or prophylactic ventilation, chronic hypercapnia.
- False negatives: unmeasured ABGs or pulse-ox gaps in severely ill patients.

Pseudocode:

```
resp_score = oxygenation + vent_support + ventilatory + resp_distress
LAT_RESP_FAILURE = int(resp_score >= 2)
```

```
flowchart TD
A[Oxygenation domain] --> S
B[Mechanical ventilation] --> S
C[Hypercapnia/acidemia] --> S
D[Tachypnea] --> S
S[resp_score = sum(domains)] --> T{>= 2?}
T -->|Yes| Y[LAT_RESP_FAILURE = 1]
T -->|No| N[LAT_RESP_FAILURE = 0]
```

3) LAT_RENAL_DYSFUNCTION

- Binary label: 1 = evidence of clinically relevant renal dysfunction.
- Inputs: Creatinine, BUN, Urine, K, HCO₃, admission Weight.
- Domain indicators:
 - D1_filtration = (max(Creatinine) >= 2.0) OR (max(BUN) >= 40)
 - D2_oliguria = urine_24h < 500 OR urine_mlgph < 0.5
 - D3_renal_metabolic = (max(K) >= 5.5) OR (min(HCO₃) < 18)
- Score: renal_score = D1 + D2 + D3
- Threshold: LAT_RENAL_DYSFUNCTION = 1 if renal_score >= 2 else 0
- Missingness: if urine unavailable, do not assume adequate output; if only one domain is observed, set NA.
- False positives: chronic kidney disease, diuretic-related low urine.
- False negatives: early AKI before creatinine rises.

Pseudocode:

```
renal_score = filtration + oliguria + renal_metabolic
LAT_RENAL_DYSFUNCTION = int(renal_score >= 2)
```

```
flowchart TD
A[Creatinine/BUN domain] --> S
B[Oliguria domain] --> S
C[Potassium/HCO3 domain] --> S
S[renal_score = sum(domains)] --> T{>= 2?}
T -->|Yes| Y[LAT_RENAL_DYSFUNCTION = 1]
T -->|No| N[LAT_RENAL_DYSFUNCTION = 0]
```

4) LAT_HEPATIC_DYSFUNCTION

- Binary label: 1 = evidence of hepatic dysfunction.
- Inputs: Bilirubin, AST, ALT, ALP, Albumin.
- Domain indicators:
 - D1_cholestasis = max(Bilirubin) >= 2.0
 - D2_hepatocellular = (max(AST) >= 150) OR (max(ALT) >= 150) OR (max(ALP) >= 300)
 - D3_synthetic_proxy = min(Albumin) < 2.5
- Score: hepatic_score = D1 + D2 + D3
- Threshold: LAT_HEPATIC_DYSFUNCTION = 1 if hepatic_score >= 2 else 0
- Missingness: if bilirubin and transaminases are all absent, set NA; albumin alone is insufficient.
- False positives: chronic liver disease, malnutrition, hemodilution.
- False negatives: hepatic dysfunction not tested in first window.

Pseudocode:

```
hepatic_score = cholestasis + hepatocellular + synthetic_proxy
LAT_HEPATIC_DYSFUNCTION = int(hepatic_score >= 2)
```

```
flowchart TD
A[Bilirubin domain] --> S
B[AST/ALT/ALP domain] --> S
C[Albumin domain] --> S
S[hepatic_score = sum(domains)] --> T{>= 2?}
T -->|Yes| Y[LAT_HEPATIC_DYSFUNCTION = 1]
T -->|No| N[LAT_HEPATIC_DYSFUNCTION = 0]
```

5) LAT_NEURO_DYSFUNCTION

- Binary label: 1 = clinically meaningful depressed consciousness proxy.
- Inputs: GCS.
- Domain indicators:
 - D1_moderate = min(GCS) < 13
 - D2_severe = min(GCS) <= 8
- Score: neuro_score = D1 + D2
- Threshold: LAT_NEURO_DYSFUNCTION = 1 if neuro_score >= 1 else 0
- Missingness: if all GCS missing, NA.
- False positives: sedation, intubation-related charting artifacts.
- False negatives: delirium without low GCS.

Pseudocode:

```
neuro_score = moderate_gcs + severe_gcs
LAT_NEURO_DYSFUNCTION = int(neuro_score >= 1)
```

```
flowchart TD
A[GCS < 13] --> S
B[GCS <= 8] --> S
```

```

S[neuro_score = sum(domains)] --> T{>= 1?}
T -->|Yes| Y[LAT_NEURO_DYSFUNCTION = 1]
T -->|No| N[LAT_NEURO_DYSFUNCTION = 0]

```

6) LAT_METABOLIC_DERANGEMENT

- Binary label: 1 = clinically meaningful acid-base or electrolyte/metabolic instability.
- Inputs: pH, HCO₃, Lactate, Glucose, Na, K, Mg.
- Domain indicators:
 - D1_acidemia = min(pH) < 7.30
 - D2_bicarbonate = min(HCO₃) < 18
 - D3_lactate = max(Lactate) > 2.2
 - D4_glucose = (max(Glucose) > 250) OR (min(Glucose) < 60)
 - D5_electrolytes = (max(Na) > 150) OR (min(Na) < 130) OR (max(K) > 5.5) OR (min(K) < 3.0) OR (max(Mg) > 3.0) OR (min(Mg) < 1.2)
- Score: `met_score = D1 + D2 + D3 + D4 + D5`
- Threshold: `LAT_METABOLIC_DERANGEMENT = 1 if met_score >= 2 else 0`
- Missingness: require at least 2 metabolic domains observed.
- False positives: isolated diabetes dysglycemia, chronic electrolyte disorder.
- False negatives: occult metabolic stress with absent lactate or ABG.

Pseudocode:

```

met_score = acidemia + bicarbonate + lactate + glucose + electrolytes
LAT_METABOLIC_DERANGEMENT = int(met_score >= 2)

```

```

flowchart TD
A[pH domain] --> S
B[HCO3 domain] --> S
C[Lactate domain] --> S
D[Glucose domain] --> S
E[Electrolyte domain] --> S
S[met_score = sum(domains)] --> T{>= 2?}
T -->|Yes| Y[LAT_METABOLIC_DERANGEMENT = 1]
T -->|No| N[LAT_METABOLIC_DERANGEMENT = 0]

```

7) LAT_GLOBAL_SEVERITY

- Binary label: 1 = multisystem physiologic instability.
- Inputs: previously derived latent tags.
- Domain indicators:
 - D1 = LAT_SHOCK
 - D2 = LAT_RESP_FAILURE
 - D3 = LAT_RENAL_DYSFUNCTION
 - D4 = LAT_HEPATIC_DYSFUNCTION
 - D5 = LAT_NEURO_DYSFUNCTION
 - D6 = LAT_METABOLIC_DERANGEMENT
- Score: `global_score = D1 + D2 + D3 + D4 + D5 + D6`
- Threshold: `LAT_GLOBAL_SEVERITY = 1 if global_score >= 2 else 0`
- Missingness: compute only if at least 4 constituent latents are evaluable.
- False positives: chronic stable multisystem disease.
- False negatives: catastrophic single-organ failure.

Pseudocode:

```

global_score = shock + resp + renal + hepatic + neuro + metabolic
LAT_GLOBAL_SEVERITY = int(global_score >= 2)

```

```

flowchart TD
A[LAT_SHOCK] --> S
B[LAT_RESP_FAILURE] --> S
C[LAT_RENAL_DYSFUNCTION] --> S
D[LAT_HEPATIC_DYSFUNCTION] --> S
E[LAT_NEURO_DYSFUNCTION] --> S
F[LAT_METABOLIC_DERANGEMENT] --> S
S[global_score = sum(latents)] --> T{>= 2?}
T -->|Yes| Y[LAT_GLOBAL_SEVERITY = 1]
T -->|No| N[LAT_GLOBAL_SEVERITY = 0]

```

Observed-to-latent measurement model

Observed variable	Parent latent(s)	Abnormal direction	Specificity	Used in tree?	Notes
Albumin [^2]	LAT_HEPATIC_DYSFUNCTION, LAT_GLOBAL_SEVERITY	Lower worse	Non-specific	Yes	Low albumin is chronicity/nutrition/hemodilution sensitive.
ALP [^2]	LAT_HEPATIC_DYSFUNCTION	Higher worse	Moderate	Yes	More cholestatic than synthetic.
ALT [^2]	LAT_HEPATIC_DYSFUNCTION	Higher worse	Moderate	Yes	Hepatocellular injury proxy.
AST [^2]	LAT_HEPATIC_DYSFUNCTION	Higher worse	Moderate	Yes	Can rise with shock liver or muscle injury.
Bilirubin [^2]	LAT_HEPATIC_DYSFUNCTION	Higher worse	Relatively specific	Yes	Best hepatic latent anchor in this dataset.
BUN [^2]	LAT_RENAL_DYSFUNCTION, LAT_GLOBAL_SEVERITY	Higher worse	Moderate	Yes	Volume status also matters.
Cholesterol [^2]	None retained	Lower often worse	Weak	No	Too sparse for final DAG role [^1].
Creatinine [^2]	LAT_RENAL_DYSFUNCTION	Higher worse	Relatively specific	Yes	Core renal indicator.
DiasABP [^2]	LAT_SHOCK, LAT_GLOBAL_SEVERITY	Lower worse	Moderate	Indirectly	Backup BP signal.
FIO2 [^2]	LAT_RESP_FAILURE, MISS_CARE_INTENSITY	Higher worse	Non-specific	Yes	Part physiologic need, part clinician response.
GCS [^2]	LAT_NEURO_DYSFUNCTION, LAT_GLOBAL_SEVERITY	Lower worse	High	Yes	Near-direct but still confounded by sedation.
Glucose [^2]	LAT_METABOLIC_DERANGEMENT, LAT_GLOBAL_SEVERITY	Both low/high worse	Non-specific	Yes	Stress and treatment responsive.
HCO3 [^2]	LAT_METABOLIC_DERANGEMENT, LAT_RENAL_DYSFUNCTION, LAT_SHOCK	Lower worse	Moderate	Yes	Shared child of several latents.
HCT [^2]	LAT_GLOBAL_SEVERITY	Lower worse	Non-specific	No	Bleeding, dilution, chronic anemia all possible.
HR [^2]	LAT_SHOCK, LAT_GLOBAL_SEVERITY	Higher worse	Non-specific	Yes	Also affected by pain, fever, drugs.
K [^2]	LAT_RENAL_DYSFUNCTION, LAT_METABOLIC_DERANGEMENT	Both low/high worse	Moderate	Yes	Renal and metabolic cross-load.
Lactate [^2]	LAT_SHOCK, LAT_METABOLIC_DERANGEMENT, LAT_GLOBAL_SEVERITY	Higher worse	Moderate	Yes	Strong hypoperfusion marker when measured.
Mg [^2]	LAT_METABOLIC_DERANGEMENT, LAT_RENAL_DYSFUNCTION	Both low/high worse	Weak-moderate	Yes	Supportive only.
MAP [^2]	LAT_SHOCK, LAT_GLOBAL_SEVERITY	Lower worse	High for hemodynamics	Yes	Best pressure summary.
MechVent [^2]	LAT_RESP_FAILURE, MISS_CARE_INTENSITY	1 usually worse	Mixed state/treatment	Yes	Treat as care-process descendant, not pure biology.

Observed variable	Parent latent(s)	Abnormal direction	Specificity	Used in tree?	Notes
Na [^2]	LAT_METABOLIC_DERANGEMENT, LAT_RENAL_DYSFUNCTION	Both low/high worse	Moderate	Yes	Shared metabolic child.
NIDiasABP [^2]	LAT_SHOCK	Lower worse	Moderate	Indirectly	Use when invasive BP absent.
NIMAP [^2]	LAT_SHOCK	Lower worse	High	Yes	Use when invasive MAP absent.
NISysABP [^2]	LAT_SHOCK	Lower worse	High	Yes	Use when invasive SBP absent.
PaCO2 [^2]	LAT_RESP_FAILURE, LAT_METABOLIC_DERANGEMENT	Higher worse	Moderate	Yes	Ventilatory failure marker.
PaO2 [^2]	LAT_RESP_FAILURE	Lower worse	High	Yes	Most useful with FiO2.
pH [^2]	LAT_METABOLIC_DERANGEMENT, LAT_RESP_FAILURE, LAT_SHOCK	Lower worse	Shared	Yes	Generic severity indicator.
Platelets [^2]	LAT_GLOBAL_SEVERITY	Lower worse	Weak-moderate	No	Optional hematologic latent was not retained.
RespRate [^2]	LAT_RESP_FAILURE, LAT_GLOBAL_SEVERITY	Higher worse	Non-specific	Yes	Sparse in this dataset [^1].
SaO2 [^2]	LAT_RESP_FAILURE	Lower worse	Moderate	Yes	Pulse-ox proxy when ABG absent.
SysABP [^2]	LAT_SHOCK	Lower worse	High	Yes	Core shock measurement.
Temp [^2]	LAT_GLOBAL_SEVERITY	Both low/high worse	Non-specific	No	Infection burden not identifiable as final latent.
TropI [^2]	None retained	Higher worse	Specific but too sparse	No	Cardiac latent rejected [^1].
TropT [^2]	None retained	Higher worse	Specific but sparse	No	Cardiac latent rejected [^1].
Urine [^2]	LAT_RENAL_DYSFUNCTION, LAT_SHOCK	Lower worse	Moderate	Yes	Shared renal-perfusion output.
WBC [^2]	LAT_GLOBAL_SEVERITY	Both low/high worse	Non-specific	No	Supports optional inflammation concept only.
Weight time series [^2]	MISS_CARE_INTENSITY, LAT_GLOBAL_SEVERITY	Rising may be worse	Weak	No	Better treated as process/volume proxy than core latent indicator.

Implementation specification

- Required raw columns: all background descriptors plus the time-series variables listed above from the challenge files.[^2]
- Derived features per patient and per index window: first, min, max, mean, count, missing_flag, time_of_first, time_of_worst, urine_24h, urine_m1kgph, and pf_ratio = PaO2 / FiO2 when both are observed.
- One function per latent tag: tag_shock(), tag_resp_failure(), tag_renal(), tag_hepatic(), tag_neuro(), tag_metabolic(), tag_global_severity().
- Expected output columns: LAT_SHOCK, LAT_RESP_FAILURE, LAT_RENAL_DYSFUNCTION, LAT_HEPATIC_DYSFUNCTION, LAT_NEURO_DYSFUNCTION, LAT_METABOLIC_DERANGEMENT, LAT_GLOBAL_SEVERITY.
- Missing-variable policy: return NA when fewer than the minimum number of domains are observed; do not impute normality for absent lactate, ABG, liver tests, or GCS.
- Unit assumptions: use official challenge units exactly as specified in the dataset page.[^2]
- Validation checks: reject impossible negatives except official -1 missing code, collapse invasive and non-invasive BP into common BP summaries, verify FiO2 is in 0–1 scale, and ensure no outcome columns enter tag functions.[^2]

DAG, mortality paths, and causal implications

The proposed DAG is sparse by design: background variables affect latent physiologic states; latent states affect mortality; observed labs/vitals are mainly children of latent states; and measurement intensity is a separate process influenced by severity and ICU context. This is more clinically plausible than a dense graph of lab-to-lab arrows, and it matches the official challenge structure of irregular repeated measurements over 48 hours with incomplete observations.^{[1][2]}

Core edge list

Source	Target	Edge type	Mechanism	Confidence	Keep?
BG_Age	LAT_GLOBAL_SEVERITY	Background-to-latent	Older patients are more vulnerable to acute physiologic collapse	High	Keep
BG_Age	OUT_INHOSP_DEATH	Background-to-outcome	Baseline mortality vulnerability not fully mediated by measured physiology	High	Keep
BG_Gender	LAT_GLOBAL_SEVERITY	Background-to-latent	Sex-related case-mix and physiologic reserve differences	Medium	Keep
BG_ICUType	LAT_GLOBAL_SEVERITY	Background-to-latent	ICU service captures case-mix and illness context	High	Keep
BG_ICUType	MISS_CARE_INTENSITY	Measurement-process	ICU type shapes what gets measured and how often	High	Keep
BG_Height	LAT_RENAL_DYSFUNCTION	Background-to-latent	Body size enters urine normalization and physiologic reserve	Low	Optional
BG_AdmWeight	LAT_RENAL_DYSFUNCTION	Background-to-latent	Needed for oliguria normalization and fluid burden context	Medium	Keep
LAT_GLOBAL_SEVERITY	LAT_SHOCK	Latent-to-latent	Severe acute illness often manifests as circulatory instability	High	Keep
LAT_GLOBAL_SEVERITY	LAT_RESP_FAILURE	Latent-to-latent	Severe illness often manifests as gas-exchange/ventilatory failure	High	Keep
LAT_GLOBAL_SEVERITY	LAT_RENAL_DYSFUNCTION	Latent-to-latent	Severe illness promotes AKI through hypoperfusion/inflammation	High	Keep
LAT_GLOBAL_SEVERITY	LAT_HEPATIC_DYSFUNCTION	Latent-to-latent	Severe illness can drive ischemic or cholestatic liver dysfunction	Medium	Keep
LAT_GLOBAL_SEVERITY	LAT_NEURO_DYSFUNCTION	Latent-to-latent	Multisystem instability can depress consciousness	High	Keep
LAT_GLOBAL_SEVERITY	LAT_METABOLIC_DERANGEMENT	Latent-to-latent	Severe illness drives acid-base and electrolyte disorder	High	Keep
LAT_SHOCK	LAT_RENAL_DYSFUNCTION	Latent-to-latent	Hypoperfusion causes oliguria and creatinine rise	High	Keep
LAT_SHOCK	LAT_HEPATIC_DYSFUNCTION	Latent-to-latent	Shock liver / ischemic hepatopathy pathway	Medium	Keep
LAT_SHOCK	LAT_METABOLIC_DERANGEMENT	Latent-to-latent	Tissue hypoperfusion raises lactate and acidosis	High	Keep
LAT_RESP_FAILURE	LAT_METABOLIC_DERANGEMENT	Latent-to-latent	CO2 retention and hypoxemia worsen acid-base status	High	Keep
LAT_RENAL_DYSFUNCTION	LAT_METABOLIC_DERANGEMENT	Latent-to-latent	AKI causes acidosis and electrolyte abnormalities	High	Keep
LAT_GLOBAL_SEVERITY	OUT_INHOSP_DEATH	Latent-to-mortality	Multisystem instability directly raises death risk	High	Keep
LAT_SHOCK	OUT_INHOSP_DEATH	Latent-to-mortality	Persistent hemodynamic failure is lethal	High	Keep

Source	Target	Edge type	Mechanism	Confidence	Keep?
LAT_RESP_FAILURE	OUT_INHOSP_DEATH	Latent-to-mortality	Severe gas-exchange or ventilatory failure is lethal	High	Keep
LAT_RENAL_DYSFUNCTION	OUT_INHOSP_DEATH	Latent-to-mortality	Organ failure contributes directly and via metabolic collapse	High	Keep
LAT_HEPATIC_DYSFUNCTION	OUT_INHOSP_DEATH	Latent-to-mortality	Hepatic failure marks high-risk multisystem deterioration	Medium	Keep
LAT_NEURO_DYSFUNCTION	OUT_INHOSP_DEATH	Latent-to-mortality	Severe encephalopathy/coma is outcome-proximal	High	Keep
LAT_METABOLIC_DERANGEMENT	OUT_INHOSP_DEATH	Latent-to-mortality	Severe acidosis/electrolyte crisis is outcome-proximal	High	Keep
LAT_SHOCK	OBS_MAP / OBS_SysABP / OBS_HR / OBS_Lactate / OBS_Urine	Latent-to-observed	Hemodynamic latent generates shock markers	High	Keep
LAT_RESP_FAILURE	OBS_PaO2 / OBS_FiO2 / OBS_SaO2 / OBS_PaCO2 / OBS_RespRate / OBS_MechVent	Latent-to-observed	Respiratory latent generates oxygenation/ventilation markers	High	Keep
LAT_RENAL_DYSFUNCTION	OBS_Creatinine / OBS_BUN / OBS_Urine / OBS_K / OBS_HCO3	Latent-to-observed	Renal latent generates renal and acid-base markers	High	Keep
LAT_HEPATIC_DYSFUNCTION	OBS_Bilirubin / OBS_AST / OBS_ALT / OBS_ALP / OBS_Albumin	Latent-to-observed	Hepatic latent generates liver lab pattern	Medium	Keep
LAT_NEURO_DYSFUNCTION	OBS_GCS	Latent-to-observed	Neuro latent manifests through consciousness score	High	Keep
LAT_METABOLIC_DERANGEMENT	OBS_pH / OBS_HCO3 / OBS_Lactate / OBS_Glucose / OBS_Na / OBS_K / OBS_Mg	Latent-to-observed	Metabolic latent generates acid-base and electrolyte pattern	High	Keep
LAT_GLOBAL_SEVERITY	MISS_CARE_INTENSITY	Measurement-process	Sicker patients are tested more often	High	Keep
LAT_RESP_FAILURE	MISS_ABG_ORDERING	Measurement-process	Respiratory failure increases ABG/FiO2 monitoring	High	Keep
LAT_RENAL_DYSFUNCTION	MISS_CHEMISTRY_ORDERING	Measurement-process	Renal concern increases chemistry and urine monitoring	High	Keep
MISS_CARE_INTENSITY	OBS_availability_*	Measurement-process	Observation occurs only if measured	High	Keep

Mortality mechanism

The dominant mortality path is background vulnerability -> LAT_GLOBAL_SEVERITY -> organ dysfunction -> OUT_INHOSP_DEATH, with age and ICU type acting as baseline risk shapers and the organ latents carrying most of the acute causal burden. A second major path is LAT_GLOBAL_SEVERITY -> LAT_SHOCK -> LAT_RENAL_DYSFUNCTION / LAT_METABOLIC_DERANGEMENT -> OUT_INHOSP_DEATH, which matches common ICU deterioration trajectories.^{[1][2]}

A third path is LAT_GLOBAL_SEVERITY -> LAT_RESP_FAILURE -> LAT_METABOLIC_DERANGEMENT -> OUT_INHOSP_DEATH, especially when oxygenation failure or CO2 retention drives acidemia. Hepatic and neurologic dysfunction are best treated as additional outcome-proximal organ states rather than baseline confounders.^{[3][2]}

Missingness and irregular sampling

The dataset documentation explicitly states that measurements may occur hourly, daily, or irregularly as required, and that not all variables are present in all patients. That makes MCAR implausible for many channels.^[^2]

ABGs, FiO2, lactate, liver tests, troponins, and even respiratory rate are much less complete than routine chemistries and blood pressures, which strongly suggests ordering driven by clinician suspicion, care intensity, and latent severity. The challenge paper also notes frequent missing observations and that some measurements can reflect intervention as much as genuine state.^[^1]

In modern ICU EHR literature, measurement frequency itself is often informative about the underlying patient state, and missingness patterns are associated with mortality and demographics. For this dataset, the most defensible assumption is a mix of MAR and MNAR driven by severity, ICU context, and clinician decisions, so missingness should enter the causal graph through explicit process nodes such as `MISS_CARE_INTENSITY`, `MISS_ABG_ORDERING`, and `MISS_CHEMISTRY_ORDERING`.^{[5][6]}

Backdoor and adjustment implications

Likely background confounders for many mortality-effect questions are `BG_Age`, `BG_Gender`, `BG_ICUType`, and sometimes body-size variables if urine-normalized renal states are involved. These precede acute physiology and are not children of ICU measurements.^[^2]

Most approved latent tags are **mediators** or outcome-proximal severity markers, not generic adjustment variables. For example, if the estimand is the total effect of respiratory failure on mortality, adjusting for `LAT_METABOLIC_DERANGEMENT`, `LAT_GLOBAL_SEVERITY`, or `LAT_RENAL_DYSFUNCTION` can block real causal pathways or induce bad-control bias.

Likely colliders to avoid conditioning on include measurement intensity, ABG ordering, and possibly `MechVent` when severity drives both respiratory failure and the decision to ventilate. Because `MechVent` is an explicit care-process variable and also a descendant of respiratory severity, it is unsafe as a routine adjustment covariate in disease-state effect estimation.^[^2]

Graph specification

Adjacency list

- `BG_Age` -> `LAT_GLOBAL_SEVERITY`, `OUT_INHOSP_DEATH`
- `BG_Gender` -> `LAT_GLOBAL_SEVERITY`
- `BG_ICUType` -> `LAT_GLOBAL_SEVERITY`, `MISS_CARE_INTENSITY`
- `BG_AdmWeight` -> `LAT_RENAL_DYSFUNCTION`
- `LAT_GLOBAL_SEVERITY` -> `LAT_SHOCK`, `LAT_RESP_FAILURE`, `LAT_RENAL_DYSFUNCTION`, `LAT_HEPATIC_DYSFUNCTION`, `LAT_NEURO_DYSFUNCTION`, `LAT_METABOLIC_DERANGEMENT`, `OUT_INHOSP_DEATH`, `MISS_CARE_INTENSITY`
- `LAT_SHOCK` -> `LAT_RENAL_DYSFUNCTION`, `LAT_HEPATIC_DYSFUNCTION`, `LAT_METABOLIC_DERANGEMENT`, `OUT_INHOSP_DEATH`
- `LAT_RESP_FAILURE` -> `LAT_METABOLIC_DERANGEMENT`, `OUT_INHOSP_DEATH`, `MISS_ABG_ORDERING`
- `LAT_RENAL_DYSFUNCTION` -> `LAT_METABOLIC_DERANGEMENT`, `OUT_INHOSP_DEATH`, `MISS_CHEMISTRY_ORDERING`
- `LAT_HEPATIC_DYSFUNCTION` -> `OUT_INHOSP_DEATH`
- `LAT_NEURO_DYSFUNCTION` -> `OUT_INHOSP_DEATH`
- `LAT_METABOLIC_DERANGEMENT` -> `OUT_INHOSP_DEATH`
- `MISS_CARE_INTENSITY` -> `OBS_*` availability
- `MISS_ABG_ORDERING` -> `OBS_PaO2`, `OBS_PaCO2`, `OBS_pH`, `OBS_FiO2`, `OBS_SaO2`
- `MISS_CHEMISTRY_ORDERING` -> `OBS_Lactate`, `OBS_Creatinine`, `OBS_BUN`, `OBS_AST`, `OBS_ALT`, `OBS_Bilirubin`, `OBS_Albumin`

Mermaid DAG

```
graph TD
  BG_Age --> LAT_GLOBAL_SEVERITY
  BG_Age --> OUT_INHOSP_DEATH
  BG_Gender --> LAT_GLOBAL_SEVERITY
  BG_ICUType --> LAT_GLOBAL_SEVERITY
  BG_ICUType --> MISS_CARE_INTENSITY
  BG_AdmWeight --> LAT_RENAL_DYSFUNCTION

  LAT_GLOBAL_SEVERITY --> LAT_SHOCK
  LAT_GLOBAL_SEVERITY --> LAT_RESP_FAILURE
  LAT_GLOBAL_SEVERITY --> LAT_RENAL_DYSFUNCTION
  LAT_GLOBAL_SEVERITY --> LAT_HEPATIC_DYSFUNCTION
  LAT_GLOBAL_SEVERITY --> LAT_NEURO_DYSFUNCTION
  LAT_GLOBAL_SEVERITY --> LAT_METABOLIC_DERANGEMENT
  LAT_GLOBAL_SEVERITY --> OUT_INHOSP_DEATH
  LAT_GLOBAL_SEVERITY --> MISS_CARE_INTENSITY

  LAT_SHOCK --> LAT_RENAL_DYSFUNCTION
  LAT_SHOCK --> LAT_HEPATIC_DYSFUNCTION
```

LAT_SHOCK --> LAT_METABOLIC_DERANGEMENT
LAT_SHOCK --> OUT_INHOSP_DEATH

LAT_RESP_FAILURE --> LAT_METABOLIC_DERANGEMENT
LAT_RESP_FAILURE --> OUT_INHOSP_DEATH
LAT_RESP_FAILURE --> MISS_ABG_ORDERING

LAT_RENAL_DYSFUNCTION --> LAT_METABOLIC_DERANGEMENT
LAT_RENAL_DYSFUNCTION --> OUT_INHOSP_DEATH
LAT_RENAL_DYSFUNCTION --> MISS_CHEMISTRY_ORDERING

LAT_HEPATIC_DYSFUNCTION --> OUT_INHOSP_DEATH
LAT_NEURO_DYSFUNCTION --> OUT_INHOSP_DEATH
LAT_METABOLIC_DERANGEMENT --> OUT_INHOSP_DEATH

LAT_SHOCK --> OBS_MAP
LAT_SHOCK --> OBS_SysABP
LAT_SHOCK --> OBS_HR
LAT_SHOCK --> OBS_Lactate
LAT_SHOCK --> OBS_Urine

LAT_RESP_FAILURE --> OBS_PaO2
LAT_RESP_FAILURE --> OBS

[^10][^11][^12][^13][^14][^15][^16][^17][^18][^19][^20][^21][^22][^23][^24][^25

<div align="center">✱✱✱</div>

[^1]: <https://archive.physionet.org/pn3/challenge/2012/>
[^2]: <https://www.cinc.org/archives/2012/pdf/0257.pdf>
[^3]: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9837980/>
[^4]: <https://www.msmanuals.com/professional/multimedia/table/sequential-organ-failure-assessment-sof>
[^5]: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12220764/>
[^6]: <https://arxiv.org/html/2402.15730v1>
[^7]: <https://www.physionet.org/content/challenge-2012/1.0.0/>
[^8]: <https://github.com/Ghadeer-Ghosheh/physionet2012-timeseries>
[^9]: <https://paperswithcode.com/dataset/physionet-challenge-2012>
[^10]: <https://physionet.org/challenge/2012/>
[^11]: <https://moody-challenge.physionet.org/2012/>
[^12]: <https://arxiv.org/html/2510.16026v1>
[^13]: http://modelai.gettysburg.edu/2020/icu/handout_py/index.html
[^14]: <https://archive.physionet.org/challenge/2012/papers/>
[^15]: <https://moody-challenge.physionet.org/2012/papers/>
[^16]: <https://royalsocietypublishing.org/doi/10.1098/rsos.220638>
[^17]: <https://github.com/Msundarv/Physionet-Cardiology-Challenge-2012>
[^18]: <https://physionet.org/content/?topic=mortality>
[^19]: <https://pubmed.ncbi.nlm.nih.gov/24678516/>
[^20]: <https://files.asprtracie.hhs.gov/documents/aspr-tracie-sofa-score-fact-sheet.pdf>
[^21]: https://en.wikipedia.org/wiki/SOFA_score
[^22]: https://www.physio-pedia.com/Sequential_Organ_Failure_Assessment_Score
[^23]: <https://www.esicm.org/wp-content/uploads/2018/03/esicmworking5-2.pdf>
[^24]: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7929579/>
[^25]: <https://medinform.jmir.org/2025/1/e79307>
[^26]: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3965265/>
[^27]: https://iris.unipa.it/retrieve/e52e0e20-9358-41f7-bcac-33fc5820c274/jama_ranzani_2025_oi_250090_1
[^28]: <https://ctlm.berkeley.edu/publications/implicit-bias-critical-care-data-factors-affecting-sampl>